



Quantification and prediction of the impact of ENSO on rainfed rice yields in Thailand

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ABSTRACT

Climate variability driven by the El Niño-Southern Oscillation (ENSO) significantly impacts rainfed rice yields in Thailand, a critical agricultural region heavily reliant on monsoon rainfall. This study quantifies and predicts the effects of ENSO-induced climate signals on rice yields using advanced artificial intelligence (AI) techniques. We employed a three-stage methodology, integrating Multiple Linear Regression (MLR) with Variance Inflation Factor (VIF) analysis to assess the relative contributions of ENSO indices and local climate variables, followed by the development of two AI models: ENSO—CropNet, a deep neural network (DNN), and an ensemble Random Forest-XGBoost (RF-XGBoost) model. The results revealed that ENSO indices, particularly NINO3 and NINO3.4, significantly reduced rice yields in several provinces, with temperature and rainfall variability playing critical roles. The ENSO—CropNet model demonstrated high predictive accuracy ($R^2 = 0.89$, MAE = 1.04, RMSE = 1.45), surpassing the RF-XGBoost model ($R^2 = 0.82$, MAE = 3.62, RMSE = 3.84). Feature importance analysis identified rainfall, minimum temperature, and ENSO indices as key predictors. The study found that ENSO-driven climate variability led to a 12 % decline in rice yields across northern provinces. The findings underscore the significant role of ENSO-induced climate variability in rainfed rice production, with AI models such as ENSO—CropNet offering highly accurate predictions. These results highlight the potential of AI techniques to enhance agricultural forecasting and resilience in climate-vulnerable regions like Thailand.

1. Introduction

Climate change is a critical issue of the 21st century, causing intense changes in global climatic systems and imposing long-term adaptations in environmental dynamics (Waqas et al., 2024; Al-Amin et al., 2021; Felkner et al., 2009). It is initiating extreme weather events (Masud et al., 2016), including droughts (Salvador et al., 2020), floods (Waqas et al., 2023), and landslides (Gariano and Guzzetti, 2016), which have far-reaching socio-economic effects (Parry et al., 2004). Among the key drivers of these climatic disturbances is the El Niño Southern Oscillation (ENSO) (Battisti and Sarachik, 1995), a natural climate phenomenon characterized by irregular oscillations in wind patterns and sea surface temperatures (SSTs) across the tropical eastern Pacific Ocean (Wang and Picaut, 2004). ENSO plays a pivotal role in global climate variability,

with its two primary phases—El Niño, marked by warming SSTs, and La Niña, characterized by cooling SSTs—lasting several months and recurring at irregular intervals of a few years (Battisti and Sarachik, 1995; An et al., 2020; Fang et al., 2022). These phases influence tropical and subtropical climate systems (An et al., 2020), causing irregular weather patterns that affect rainfed agricultural productivity (Cirino et al., 2015; Paz et al., 2012; Wang et al., 2020). The impacts of ENSO on agriculture are multifaceted and complex (Cirino et al., 2015), with its influence extending to crop yields (Paz et al., 2012), production systems (Wang et al., 2020), and export prices in major agricultural regions (Adams et al., 1999). El Niño-related droughts are often associated with less rainfall, leading to massive crop failure (YOKYING and PROM-KHAMBUT, 2024), while La Niña events, characterized by excessive rainfall, can trigger flooding and complicate agricultural management

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(Rodysill et al., 2019). These disruptions are substantial for water-dependent crops such as rice and cassava, critical for food security in many regions (Pipitpukdee et al., 2020; Jintrawet and Buddhagoon, 2011). The interplay between ENSO-driven climate variability and agricultural productivity has been documented in various studies, highlighting the vulnerability of rainfed systems to these climatic oscillations (Wang and Picaut, 2004). Existing research has focused on specific countries such as Australia (Potgieter et al., 2005; Yuan and Yamagata, 2015), China (Liu et al., 2014; Lu et al., 2024; Shuai et al., 2016), the United States (Abid et al., 2018; Allen and Karoly, 2014), India (Ahmad et al., 2011; Gandhi et al., 2016), Southeast Asia (SEA) (Felkner et al., 2009; Masud et al., 2016; Pipitpukdee et al., 2020; Jintrawet and Buddhagoon, 2011; Kyaw et al., 2023; Molle, 2024; Nounmusig, 2018), and Africa (Ngaira, 2007; Sazib et al., 2020; Shukla et al., 2020). This geographic bias has limited insights into the global-scale impacts of ENSO on agricultural production. Beyond modulating oceanic and atmospheric conditions in the Pacific, ENSO affects global climate patterns (Battisti and Sarachik, 1995; Wang and Picaut, 2004; Waqas et al., 2024), precipitation variability (Waqas et al., 2024), and agricultural productivity (Waqas et al., 2024), underscoring the critical need for its accurate analysis and predictive modeling.

In SEA, where agriculture relies on rainfall, ENSO-driven climate variability has significant implications (Waqas et al., 2024). Studies have highlighted ENSO's detrimental effects on rice and maize production in Indonesia (Kirono and Tapper, 1999; Naylor et al., 2007), with similar patterns observed in the Philippines' irrigated and rainfed rice ecosystems (Angulo et al., 2012). Countries like Malaysia, Thailand, and Cambodia have faced agricultural challenges attributable to ENSO-related climate variability (Lasco et al., 2011; Reda et al., 2015; Watanabe, 2017). In the past, Thailand has experienced climate fluctuations, with temperatures rising by 1.04–1.80 °C per century and irregular rainfall patterns significantly affecting agricultural productivity (Waqas et al., 2024). About 80 % of Thai farmland relies on rainfall (Waqas et al., 2023), so understanding its spatial and temporal variability is essential (Waqas et al., 2024). Research incorporating ENSO/IOD signals into rainfall models has improved forecasting accuracy, underscoring ENSO's relevance in regional climate prediction (Yuan and Yamagata, 2015). Monsoons influence Thailand's climate, the intertropical convergence zone, and tropical cyclones, characterized by high interannual variability (CHANALERT and Janjai, 2022; Singhrattana, 2003). This variability significantly impacts rainfed rice systems, which account for 65 % of the country's total rice-growing area (Reda et al., 2015). ENSO events exacerbate this variability by inducing extreme droughts during El Niño and severe floods during La Niña, often coinciding with critical crop-growing periods (Wikarnpapraharn and Kositsakulchai, 2010). Wongkhunkaew et al. (2020) observed that droughts impacted production areas but not necessarily yield, highlighting the importance of soil fertility and management practices (Wongkhunkaew et al., 2020). ENSO-driven droughts hinder key physiological processes such as photosynthesis, reducing grain yield by up to 5.7 % during intense events like the 2015 El Niño (Shuai et al., 2016). Conversely, La Niña-related floods may increase productivity by improving water availability and pose challenges such as methane emissions and pest outbreaks (Parry et al., 2004; Kumar et al., 2024). Yield impacts vary across Thailand, with rainfed systems, particularly in the northeast, most vulnerable to El Niño (Masud et al., 2016). Central irrigated areas are less affected due to water infrastructure (Wikarnpapraharn and Kositsakulchai, 2010). Mitigating ENSO's impacts requires climate-smart technologies, improved irrigation systems, and robust water forecasting tools to ensure the resilience of Thailand's rice sector (Pipitpukdee et al., 2020; Lee, 2021).

Recent advancements in artificial intelligence (AI), including machine learning (ML) and deep learning (DL), have revolutionized the analysis of ENSO impacts on crop production (Waqas et al., 2023; Saha and Nanjundiah, 2020; Wang et al., 2023). While providing valuable insights, traditional statistical models such as regression and ARIMA

often fail to account for ENSO's stochastic and non-linear nature (Su et al., 2018). Although effective for short-term forecasts, Dynamical models struggle to capture the complex interactions between ENSO phases and cropping systems (Waqas et al., 2024). AI models, particularly DL approaches like convolutional neural networks (CNNs) and recurrent neural networks (RNNs), address these limitations by analyzing large datasets to uncover hidden patterns and temporal dynamics (Kattenborn et al., 2021; Ciaburro and Venkateswaran, 2017). ML models like support vector machines (SVMs) and decision trees (DTs) have been utilized for yield anomaly prediction, though they often require extensive feature engineering (Manjunath and Palayyan, 2023; Kourou et al., 2015). Moreover, probabilistic AI frameworks integrated with tools like the Decision Support System for Agrotechnology Transfer (DSSAT) offer enhanced early warning systems, providing longer lead times to mitigate ENSO-related crop losses (Boer and Surmaini, 2020). Although there is a substantial body of research on this topic, comprehensive global datasets delineating the impacts of ENSO phases on crop yields remain limited.

This study attempts to bridge the gap in understanding the complex relation between ENSO phases and rainfed rice yield variability in Thailand by integrating AI-based predictive models with large-scale climate signals. While previous research has examined ENSO's influence on various agricultural systems, a comprehensive approach leveraging advanced AI techniques to assess region-specific impacts remains limited (Felkner et al., 2009; Masud et al., 2016; Pipitpukdee et al., 2020; Jintrawet and Buddhagoon, 2011; Kyaw et al., 2023; Molle, 2024; Nounmusig, 2018; Lasco et al., 2011; Reda et al., 2015; Watanabe, 2017). A newly developed AI framework improves ENSO-impacted rice yield prediction precision, which helps generate adaptive adaptations for climate resilience in Thai agriculture.

This study examines a research problem because current statistical and dynamical forecasting systems fold at capturing ENSO stochastic behavior together with its non-linear climate system modifications at regional levels. Traditional statistical prediction methods combined with regression models do not efficiently capture regional yield anomalies, so researchers need stronger data analytics solutions (Rembold et al., 2013; Basso and Liu, 2019). DL networks particularly CNNs, assist this work to properly analyze the effects of ENSO-driven climate variations on rainfed rice cultivation. The purpose of this study consists of two parts: studying the systematic correlation between ENSO phases and rainfed rice yield variability in Thailand and developing AI-based models to predict yield anomalies with improved accuracy. This research brings together ENSO climate signals at a large scale with predictive models based on AI for a methodological improvement in understanding climate risks affecting agriculture. The research employs global climate indices instead of traditional localized weather data for yield prediction to provide a more precise evaluation of ENSO effects. The implementation of AI-based predictive tools for Thailand's specific agro-climatic zone has established a vital contribution to climate-smart agriculture.

This paper is structured as follows: Section 1 presents the introduction, followed by Section 2, which outlines the methodology, including the data description, study area, and an overview of the AI-based models used. Sections 3 and 4 present the results and discussion, highlighting the model's performance and offering a comparison with existing literature. Finally, Section 5 provides the key conclusions of the study.

2. Materials and methods

The following structured three-stage methodology quantifies and predicts the crop yield reduction due to ENSO, as shown in Fig. 1. In stage I, data collection and preprocessing are related to meteorological and rice yield data, where missing data, outliers, and statistical analysis are concerned. Stage II employs MLR and VIF to assess the relative contribution of ENSO impacts after controlling for multicollinearity. The third stage involves creating and validating AI models RF-XGBoost and

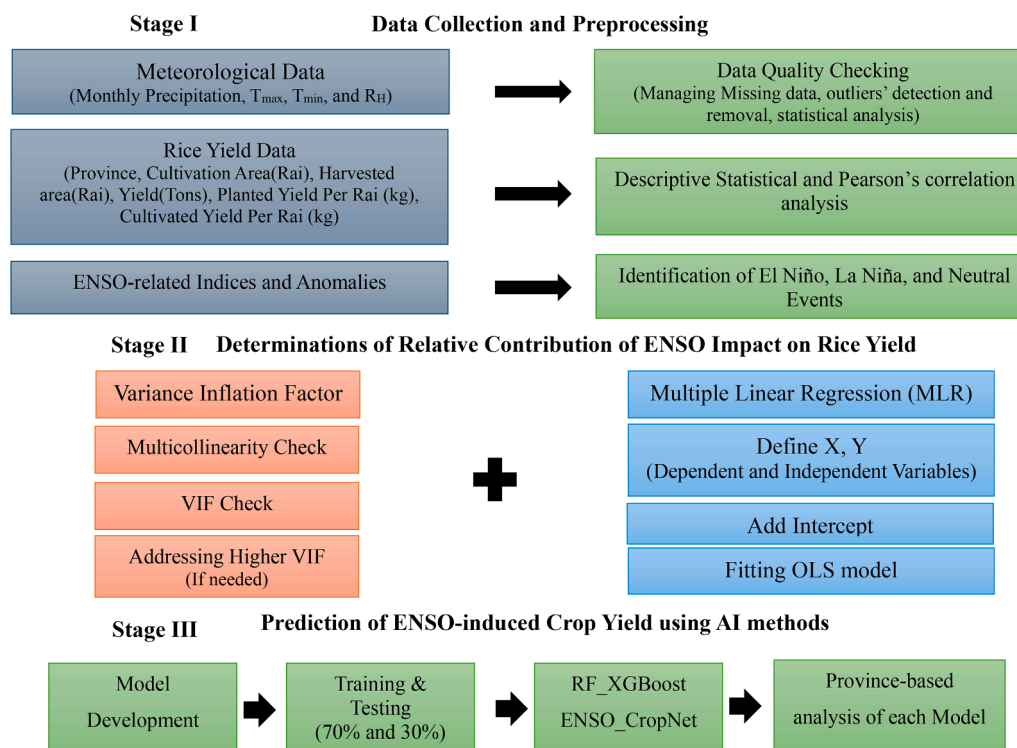


Fig. 1. Overall methodology employed to quantify and predict ENSO-Induced crop yield.

ENSO—CropNet for provincial-level analysis.

2.1. Study Region

Thailand lies between latitudes 5°37' and 20°27' N and longitudes 97°22' and 105°37' E (Waqas et al., 2024). The country is divided into five distinct climatic zones based on its meteorological characteristics and climate patterns. These regions, shaped by different topographical attributes, include the (a) central, (b) eastern, (c) northeastern, (d) northern, and (e) southern zones (M. Waqas et al., 2024). The overall climate of these regions is tropical (Waqas et al., 2024), with consistently warm temperatures throughout the year (Waqas et al., 2024). Rice cultivation, requiring about 1500 m³ of water per rai, is a critical agricultural activity in Thailand, with an average farm size of 1.6 hectares per household (Rodríguez Álvarez, 2024). Jasmine rice, the main variety, is a key export commodity despite its relatively lower yield, emphasizing Thailand's position as a leading global rice exporter (Dangthaisong et al.). The present study was conducted across all provinces of Thailand, as shown in Fig. 2.

2.2. Data Attributes and features

2.2.1. Crop data

The rice yield data from 1993 to 2023 were obtained from the Office of Agricultural Economics (OAE). The dataset includes comprehensive information on rice cultivation, harvests, and yields per rai (a traditional unit of area measurement in Thailand). This study focuses on the annual rice yield data collected from all provinces across Thailand. A summary of the descriptive statistics and correlation analysis of the historical rainy season rice yields is presented in Table 1.

2.2.2. Meteorological data

Daily meteorological data (rainfall, relative humidity, maximum and minimum temperature) for 1993–2023 were obtained from 128 weather stations distributed across Thailand from the Thai Meteorological Department (TMD). Data quality was assessed by adopting a

comprehensive methodology. LSTM-RNN model proposed by Wangwongchai et al. (2023) (Wangwongchai et al., 2023) to estimate missing values. Stations with missing data exceeding 50 % were excluded from the analysis. Given the possibility of recurring missing data due to observational or processing issues within a hydrological year, comprehensive checks were conducted to ensure data consistency. Significant deviations from expected data ranges were identified using the method of Grubbs and Beck (1972) (Grubbs and Beck, 1972). A robust quality assurance process was followed to maintain the data's accuracy, which included removing outliers and addressing any continuity issues (Zhao et al., 2018).

2.2.3. Identification of El niño, la niña, and neutral events

Previous studies have demonstrated that large-scale climate indices associated with the ENSO in the Pacific region significantly influence seasonal rainfall patterns and crop yield variability in Thailand (Felkner et al., 2009; Jintrawet and Buddhaboon, 2011; Waqas et al., 2024). Therefore, this study employs several ENSO spatial domains SST anomalies, which are assessed using Niño indices such as Niño3, Niño4, and Niño3.4 in the equatorial Pacific. In addition, other regions of interest include NIÑO3 (5°N–5°S, 150°W–90°W), NIÑO4 (5°N–5°S, 160°E–150°W), and NIÑO1+2 (0°–10°S, 90°W–80°W), all of which play a role in modulating ENSO-related climate variability. NOAA accessed the data on December 24, 2024 (<https://www.cpc.ncep.noaa.gov/data/indices/sstoi.indices>).

To identify El Niño and La Niña events from 1993 to 2023 during rice-growing season. An El Niño event was defined as an ONI >0.5 °C, while a La Niña event was characterized by an ONI value <−0.5° (Abid et al., 2018; Waqas et al., 2024). Fig. 3(a) shows the frequency of El Niño, La Niña, and Neutral events from 1993 to 2024, categorizing ENSO phases by year. Neutral events are predominant across the dataset, indicating that most years fall without extreme ENSO phases. Peaks in event frequency, such as the strong El Niño years (e.g., 1997, 2015) and significant La Niña phases (e.g., 1999, 2011), are highlighted. Fig. 3 (b) tracks anomalies (ANOM1–ANOM4) over time. Positive anomalies exceeding +0.5 identify El Niño periods, while negative anomalies

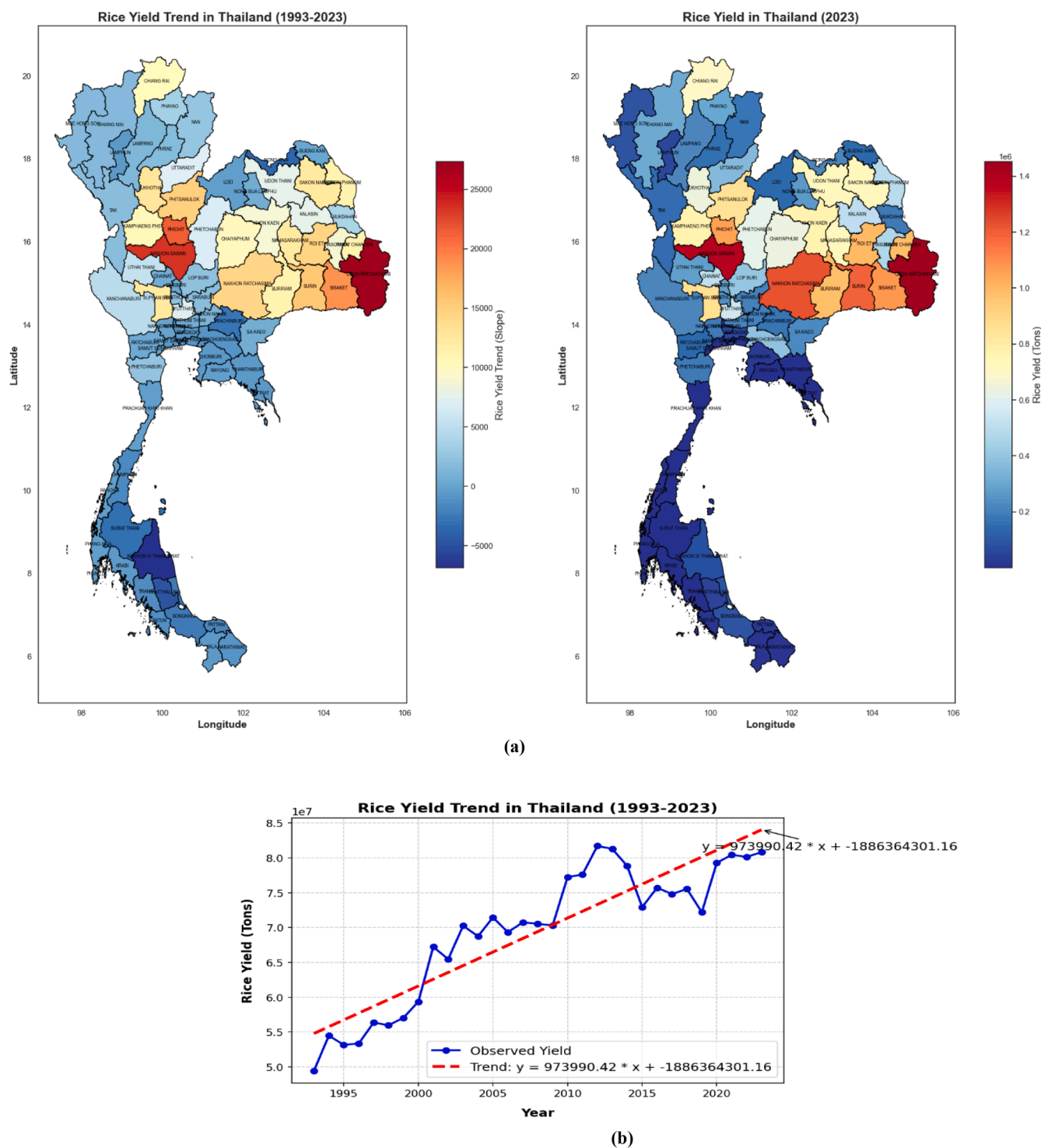


Fig. 2. (a) Major cultivation provinces and rice yield trend in Thailand, (b) Rainy Season Rice yield past trends.

below -0.5 mark La Niña periods. Neutral years align within the ± 0.5 range. Specific years and months corresponding to ENSO phases can be identified by correlating these thresholds with anomalies.

2.3. Multiple Linear regression with Variance inflation factor analysis

A Multiple Linear Regression (MLR) (Tranmer and Elliot, 2008) model combined with the Variance Inflation Factor (VIF) (Groß, 2003; O'Brien, 2007) was used to quantify the relative contributions of ENSO indices and climate variables on rice yield across different provinces.

MLR for its interpretability and ability to assess both individual and combined effects of predictors (Vaulet et al., 2022), such as ENSO indices (e.g., NINO1+2, NINO3, NINO3.4, NINO4) and climate variables (e.g., rainfall, temperature, humidity), on crop yield (Reddy et al., 2022; Aslan et al., 2024). Lobell et al. (2006) highlighted MLR's utility in studying ENSO effects on global crop yields (Lobell et al., 2006). This approach is suitable for capturing the non-linear relationships between continuous predictors and the response variable, which is essential for understanding the direct impacts of climate variability on agricultural productivity (Lobell et al., 2006). However, MLR is susceptible to

Table 1

Descriptive statistics and correlation analysis of the past rainy season rice yields.

	Planted Yield Per Rai (kg)	Cultivated Yield Per Rai (kg)	Cultivation Area(Rai)	Yield (Tons)
Mean	1030.00	1430.00	12.00	14.00
Std	120.42	120.42	1.58	1.58
Min	900.00	1300.00	10.00	12.00
Max	1200.00	1600.00	14.00	16.00
Pearson's				
Correlation				
Planted Yield Per Rai (kg)	1.00	0.83	-0.39	0.85
Cultivated Yield Per Rai (kg)	0.83	1.00	-0.53	0.98
Cultivation area (rai)	-0.39	-0.53	1.00	-0.60
Yield(tons)	0.85	0.98	-0.60	1.00

multicollinearity, a common issue in climate-agriculture studies due to the high correlation among predictors (Hou et al., 2024). To quantify multicollinearity, the reliability and interpretability of regression coefficients are ensured by incorporating VIF analysis (O'Brien, 2007). The technical description of the model is presented in Table 2. A VIF > 5 indicates significant multicollinearity, prompting the need for predictor adjustment or removal (O'Brien, 2007).

Therefore, in this study, the province-specific regression analysis enhances the model's applicability by accounting for variations in regional climate and agricultural practices. This localized approach is critical, as the impact of ENSO and climate variables on rice yield can vary across regions (Tao et al., 2006). For example, coastal provinces face intense rainfall patterns compared to inland areas, influencing the magnitude and direction of ENSO effects (Allen and Karoly, 2014; Dang et al., 2020). The model provides tailored insights for targeted interventions by isolating these regional differences. For example, a positive coefficient for NINO3.4 may indicate that El Niño events enhance rice yield in certain provinces due to favorable rainfall, while a negative coefficient for $T_{\max}$ could reflect the detrimental effects of heat stress on crop growth (Peng et al., 2004).

2.4. Prediction Models

Recent developments in DL techniques have improved the predictive

capabilities used in climate-based agricultural predictions (Hammer et al., 2001; Dhanke et al., 2024). The combination of DL techniques becomes necessary because traditional statistical and numerical models do not sufficiently handle high-dimensional and non-linear system interactions (Rembold et al., 2013; Basso and Liu, 2019). DNNs provide strong capabilities in predictive modeling solutions through their application for uniting multiple data sources (Yuan et al., 2024). Hyperspectral image classification has achieved better spatial-spectral classification accuracy through the implementation of Graph Attention Network (GAT)-based feature fusion methods that advance this field related to climate and agricultural remote sensing (Cai et al., 2024). This methodology shows potential for application in ENSO-related climate modeling because the challenging spatiotemporal patterns control crop yield responses.

The implementation of local similarity-based spatial-spectral fusion techniques has enhanced hyperspectral image classification through a solid method for selecting features and reducing dimensions (Bhatti et al., 2021). The approach utilizes methods that protect spatial data details before combining spectral data to achieve performance enhancement in classification tasks. Such forecasting models experience

Table 2

Technical description of the MLR and VIF.

Multiple Linear Regression	Variance Inflation Factor
Mathematically, $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \beta_8 X_8$ Here, Y is rice yield (tons). β_0 is the expected rice yield when all independent variables are zero. Regression coefficients (β_1 to β_8) quantify the impact of each independent variable on rice yield. Independent variables include SST anomalies in various NINO regions (X_1 to X_4), which track ENSO events, along with climate variables such as annual rainfall (X_5), maximum (X_6) and minimum temperature (X_7), and relative humidity (X_8), all influencing crop growth. (c) error term accounts for discrepancies between the observed and predicted yields.	Mathematically, $VIF_i = \frac{1}{(1 - R_i^2)}$ Here, R_i^2 Coefficient of determination was obtained by regressing X_i (i^{th} independent variable) against all other independent variables in the model. VIF_i is the Variance Inflation Factor for the i^{th} predictor. A high VIF (typically > 5) indicates X_i is highly correlated with other predictors, suggesting multicollinearity.

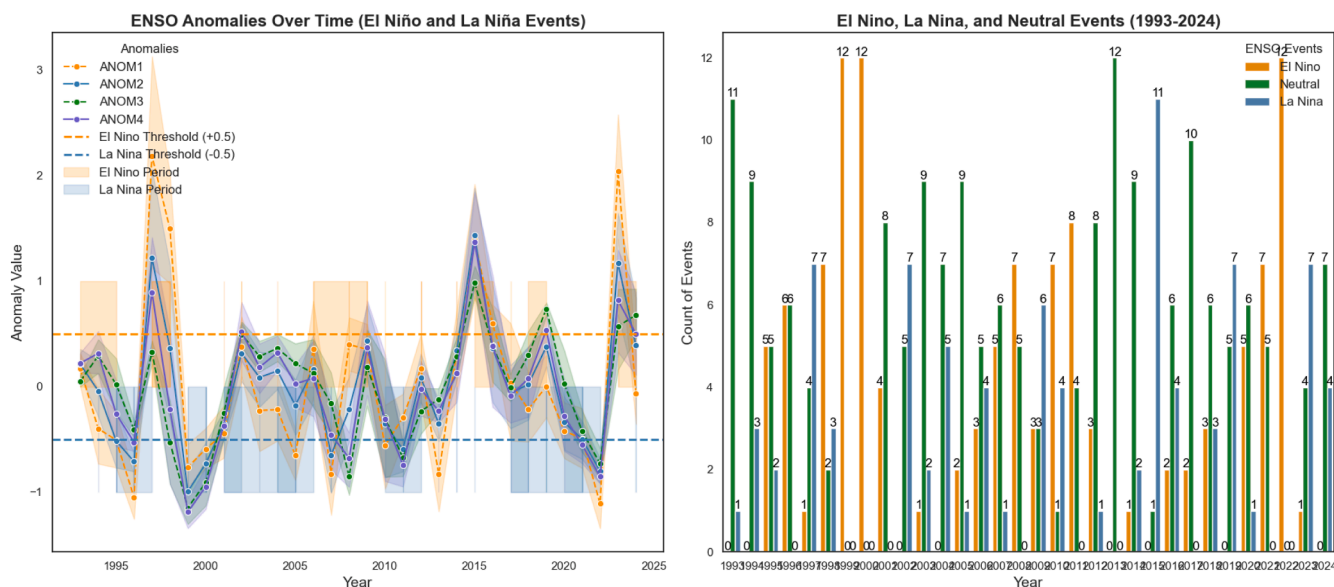


Fig. 3. ENSO anomalies over time (a) Temporal variation of ENSO anomalies (ANOM1-ANOM4) from 1993 to 2024 (b) Annual frequency of El Niño, La Niña, and Neutral events from 1993 to 2024.

difficulty when operated on extensive agricultural regions because their complexity increases alongside their computational needs (Li et al., 2019). The predictive power improvement from DL approaches requires optimization solutions to find proper computational load balance because of its extensive requirements when working with high-resolution climate data (Devaraj et al., 2021).

The selection of hyperparameters through optimization processes receives significant influence from socio-economic factors that affect AI-based agricultural forecasting (Assimakopoulos et al., 2024). Research now shows that predictive models for ENSO impact assessments require the inclusion of socio-economic variables such as land-use patterns, market access and policy interventions (Waqas et al., 2024; Fischer et al., 2005). The factors shape both agricultural data collection and the quality of information, which subsequently impacts the selection and optimization of hyperparameters. Social and economic conditions restrict data access, which requires dynamic IIP selection systems that utilize sector-specific info to achieve dependable forecasting results (Mythili and Rangaraj, 2021). The aspect takes on critical importance in ENSO-driven yield forecasting because socio-economic disruptions, including price volatility and labor shortages, reduce agricultural resilience (Whitfield et al., 2019).

The implementation of DL-based feature fusion methods optimized the hyperparameter choices for ENSO-driven rainfed crop yield forecasting. Through research improvements of conventional predictive models and solutions for computational challenges in DNN-based methods, this study improves the accuracy and application of AI-driven agricultural forecasting to support better climate adaptation strategies.

2.4.1. ENSO—CropNet

DNNs are designed to capture model complex, non-linear relationships in high-dimensional datasets (Liu et al., 2017; Liu et al., 2017). DNNs showed effectiveness in applications involving intricate data interactions, such as crop yield prediction influenced by climatic variables and large-scale climatic phenomena like the ENSO (Fang et al., 2022; Wang et al., 2023; Oikonomidis et al., 2023). Traditional ML and

statistical models often fall short of representing such intricate relationships due to their reliance on linear assumptions (Chambers and Hastie, 2017; Zhou et al., 2017). In climate-induced crop yield variability, DNNs provide a stable method for considering the temporal-spatial correlation and coupling between climate factors (Lu et al., 2024).

This study introduces ENSO—CropNet, a deep DNN-based predictive model that aims to predict rice yield in Thailand using climatic variables, ENSO indices, and lagged yield data (Fig. 4). The architecture of ENSO—CropNet consists of 4 fully connected hidden layers with progressively decreasing neurons: The number of subnets is 256, 128, 64, and 32, respectively. This architecture is designed to have a good trade-off between model complexity and computational requirements. To solve the vanishing gradient problem and improve learning for negative inputs, the Leaky ReLU activation function, Batch normalization, is used after each dense layer to reduce internal covariate shifts, stabilize training, and improve faster convergence (Ioffe, 2015) in Thailand by employing climatic variables, ENSO indices, and lagged yield data (Fig. 4).

The architecture of ENSO—CropNet consists of five fully connected hidden layers with progressively decreasing neurons: 256, 128, 64, and 32, respectively. This architecture is optimized to balance model complexity and computational efficiency. To address the vanishing gradient problem and enhance learning for negative inputs, the Leaky ReLU activation function Batch normalization is applied after each dense layer to mitigate internal covariate shifts, stabilize the training process, and accelerate convergence (Ioffe, 2015). The model is regularized by adding dropout layers with a rate of 0.3, as proposed by Srivastava et al. (2014) (Srivastava et al., 2014). The output layer has only one neuron with a linear activation function to predict the log-transformed value of rice yield. The model training uses the Adam optimizer (Kingma, 2014) with a learning rate and high computational efficiency adapted during the training process. There is gradient clipping to prevent the gradients from exploding, thus making the training process stable. Specifically, we set the learning rate schedule to a formula of warmup phase and exponential decay as the complete hyperparameters

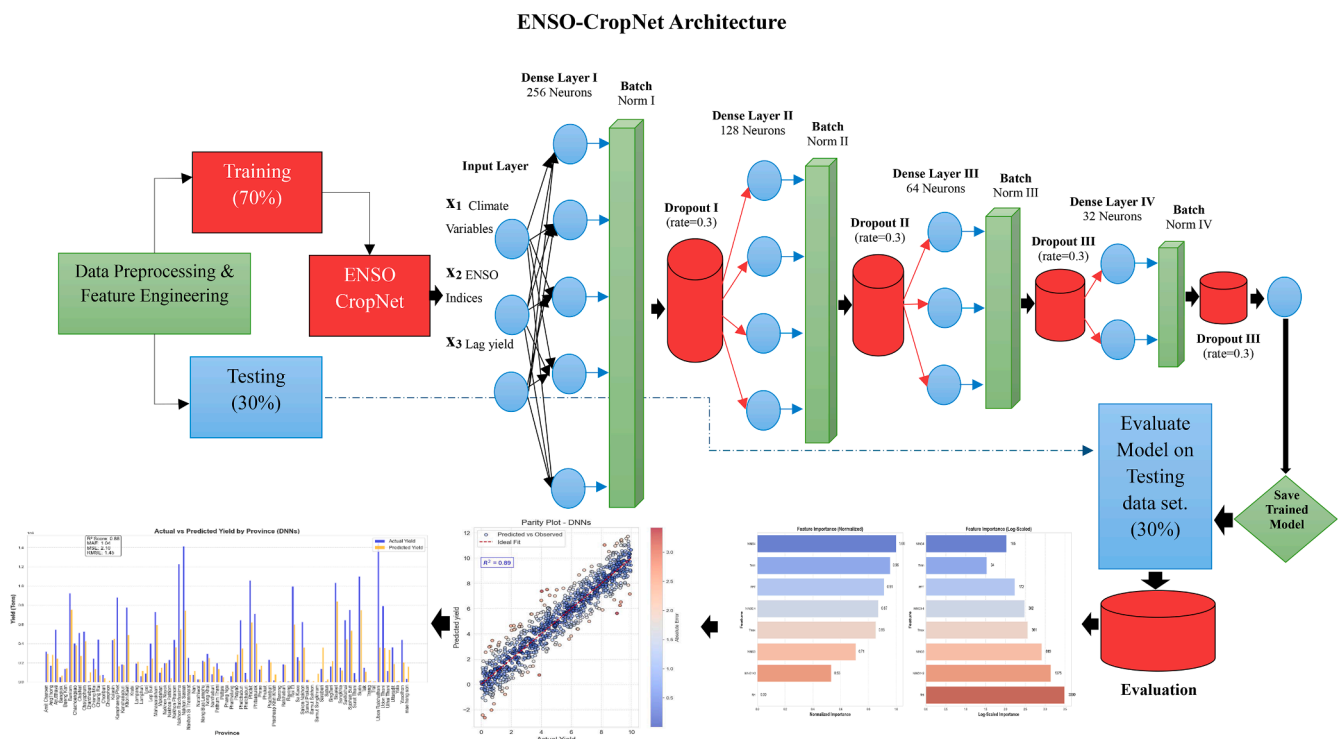


Fig. 4. Working mechanism and Architecture of the ENSO—CropNet model.

in Table 3. The hierarchical feature representation made possible by DNNs addresses the complexity of crop yield predictions due to ENSO by incorporating local and large-scale dependencies. It also explains why DNNs are well-equipped to address large, noisy datasets that are standard in agricultural research. ENSO—CropNet also resolves the problem of local and global interconnections, which is crucial in managing the complexity of crop yield predictions because of ENSO.

2.4.2. Random forest XGBoost model (RF-XGBoost)

This study used RF (Breiman, 2001) and XGBoost (Chen and Guestrin, 2016) as the ensemble model to model the climatic variables, lagged yield, and rice yield. RF and XGBoost are ensemble learning algorithms that increase prediction accuracy by creating several base models that better capture the complex interactions between predictor and outcome variables (Breiman, 2001; Friedman, 2001).

These models were chosen because they were relatively stable, could work with high dimensional data, and had been used effectively in agricultural yield prediction problems (Friedman, 2001). RF builds several decision trees during the training phase, and the final prediction is an average of the trees (Breiman, 2001). It reduces overfitting by resampling the data into subsets and choosing random features in each split, which improves model extrapolation. It benefits RF's ability to manage mixed data types, missing values, and feature importance (Liaw, 2002). The number of trees and the maximum depth of the trees were the two hyperparameters that were tuned using RandomizedSearchCV. XGBoost, on the other hand, builds trees sequentially, where each tree aims to correct the mistakes made by the previous tree using gradient boosting to optimize the loss function (Chen and Guestrin, 2016). XGBoost is fast, scalable, and efficient in accommodating large datasets, which is common in agricultural applications (Chen and Guestrin, 2016).

Hyperparameters were adjusted for accurate prediction, including learning rate, tree depth, and n_estimators (Table 4). Moreover, XGBoost provides feature importance scores that assist in interpreting the predictor's contributions. The predictors included in the dataset were climatic variables: rainfall, temperature, ENSO indices, and lagged yield (Log_Yield), while the target variable was the log-transformed rice yield (log_Yield). Model performance was evaluated using R², MAE, MSE, and RMSE (Provided in Supplementary Files), with both models demonstrating strong predictive capability. Residual and parity plots validated the models' accuracy, while SHAP identified rainfall, temperature, and lagged yield as key predictors (Lundberg, 2017). This model's ensemble nature ensures robustness against overfitting, making it ideal for predicting agricultural yields in noisy datasets.

2.4.3. Input dataset

The research uses two prediction systems, RF-XGBoost and ENSO—CropNet, to process climatic and ENSO-related variables as model inputs. The prediction models use daily meteorological data,

Table 4

Hyperparameter tuning for optimal parameter selection.

Model	Hyperparameter	Range	Selected param
RF	n_estimators	[100, 200, 300, 400]	300
	max_depth	[None, 10, 20, 30, 40, 50]	20
	min_samples_split	[2, 5, 10]	10
	min_samples_leaf	[1, 2, 4]	2
	max_features	['auto', 'sqrt', 'log2']	'sqrt'
	bootstrap	[True, False]	True
XGBoost	n_estimators	[100, 200, 300, 500]	300
	max_depth	[3, 6, 9, 12]	6
	learning_rate	[0.01, 0.05, 0.1, 0.2]	0.1
	subsample	[0.7, 0.8, 0.9, 1]	0.8
	colsample_bytree	[0.6, 0.7, 0.8, 1]	0.8
	gamma	[0, 0.1, 0.2, 0.3]	0.1
	reg_alpha	[0, 0.01, 0.1, 1]	0.1
	reg_lambda	[0, 0.1, 0.5, 1]	0.1

ENSO indices, and lagged rice yield data as input variables. The yield data from 1993 to 2023 was obtained by both the Thai Meteorological Department (TMD) and the Office of Agricultural Economics (OAE).

RF-XGBoost applies ensemble learning through multiple decision trees that combine predictions by averaging their output values. As a result, the model produces log-transformed rice yield (log_Yield). The model's prediction accuracy improves by optimizing learning rate parameters, tree depth parameters, and number of estimators parameters. The model achieves evaluation through R² measurements and RMSE and MAE metrics.

The deep neural network model ENSO—CropNet utilizes fully connected layers to learn non-linear patterns while it applies Batch normalization along with LeakyReLU activation and dropout layers. The model estimates rice yield before applying linear transformation through its activation function. The two models provide an important understanding of climate effects on yield, which helps farmers develop management plans across different ENSO phases.

Additionally, this methodology centers on evaluating how climate variability affects agriculture through forecasting models in regions that face climate change risks. The combination of weather models with DL produces more dependable climate forecasts which farmers need to make essential adjustments to their agricultural methods. Better predictions that accurately determine current weather conditions help farmers schedule planting times, determine when to irrigate, and determine what season is best for harvesting, thus reducing crop damage from weather disasters like droughts and floods.

This study enables policymakers to develop better climate adaptation strategies through its findings. The research identifies important factors that affect forecasting accuracy to uncover improvement possibilities within current weather prediction systems. The research enables government officials to deploy their resources effectively, which minimizes food security degradation caused by climate change. The outcomes from this study expand scientific understanding of the integration of AI within climate science and help future researchers in climate-smart agriculture. These improvements will maintain agricultural productivity over the long term through sustainable development that reduces climate-change-related socio-economic impacts on vulnerable populations, mainly in developing areas.

3. Results and discussion

3.1. Relative contributions of ENSO-induced climate signals to rice yield

The analysis of the relative contribution of ENSO-induced climate signals and other climate variables on rice yield across Thailand's provinces using MLR and VIF provides critical insights into the relationships between these variables and rainfed rice productivity. The results showed the influence ENSO indices exert and local climate variables PPT, T_{max}, T_{min}, and R_H on rice yield. Tables S1-S3 provide

Table 3

The hyperparameters for the optimal parameter's selection for ENSO—CropNet.

Hyperparameter	Grid Search	Optimal Value
Number of Hidden Layers	[2, 3, 4, 5]	4
Neurons per Layer	[64, 128, 256, 512]	[256, 128, 64, 32]
Activation Function	['ReLU', 'LeakyReLU', 'ELU']	LeakyReLU ($\alpha=0.1$)
Dropout Rate	[0.2, 0.3, 0.4, 0.5]	0.3
Batch Size	[32, 64, 128]	64
Learning Rate	[0.0001, 0.001, 0.01]	0.001
L2 Regularization	[0.0001, 0.001, 0.01]	0.001
Optimizer	['Adam', 'RMSprop', 'SGD']	Adam
Epochs	[100, 200, 300]	300
Early Stopping Patience	[20, 30, 40]	30
ReduceLROnPlateau Factor	[0.1, 0.2, 0.5]	0.2
ReduceLROnPlateau Patience	[5, 10, 15]	10

detailed R^2 , p-values, and VIF results, and Fig. 5(a-h) details relative contributions.

The impact of climate and ENSO indices on rice yield varies significantly across different provinces in Thailand (Table S1 and S2). In Chiang Mai, the model explains 47 % of the variance in rice yield ($R^2 = 0.470$). ENSO indices NINO3 and NINO3.4 exhibit significant negative correlations with rice yield (p-values < 0.05), suggesting that El Niño-

induced droughts reduce productivity. Additionally, T_{max} and T_{min} negatively affect yields (p-values < 0.05), while PPT is not significant (p-value = 0.034), reflecting irrigation reliance. In Khon Kaen, the model accounts for 50 % of yield variance ($R^2 = 0.500$). NINO3.4 has a significant negative impact (p-value < 0.05) alongside Tmax and Tmin (p-values < 0.01), emphasizing the detrimental effects of elevated temperatures. Due to the semi-arid climate, PPT is non-significant (p-value

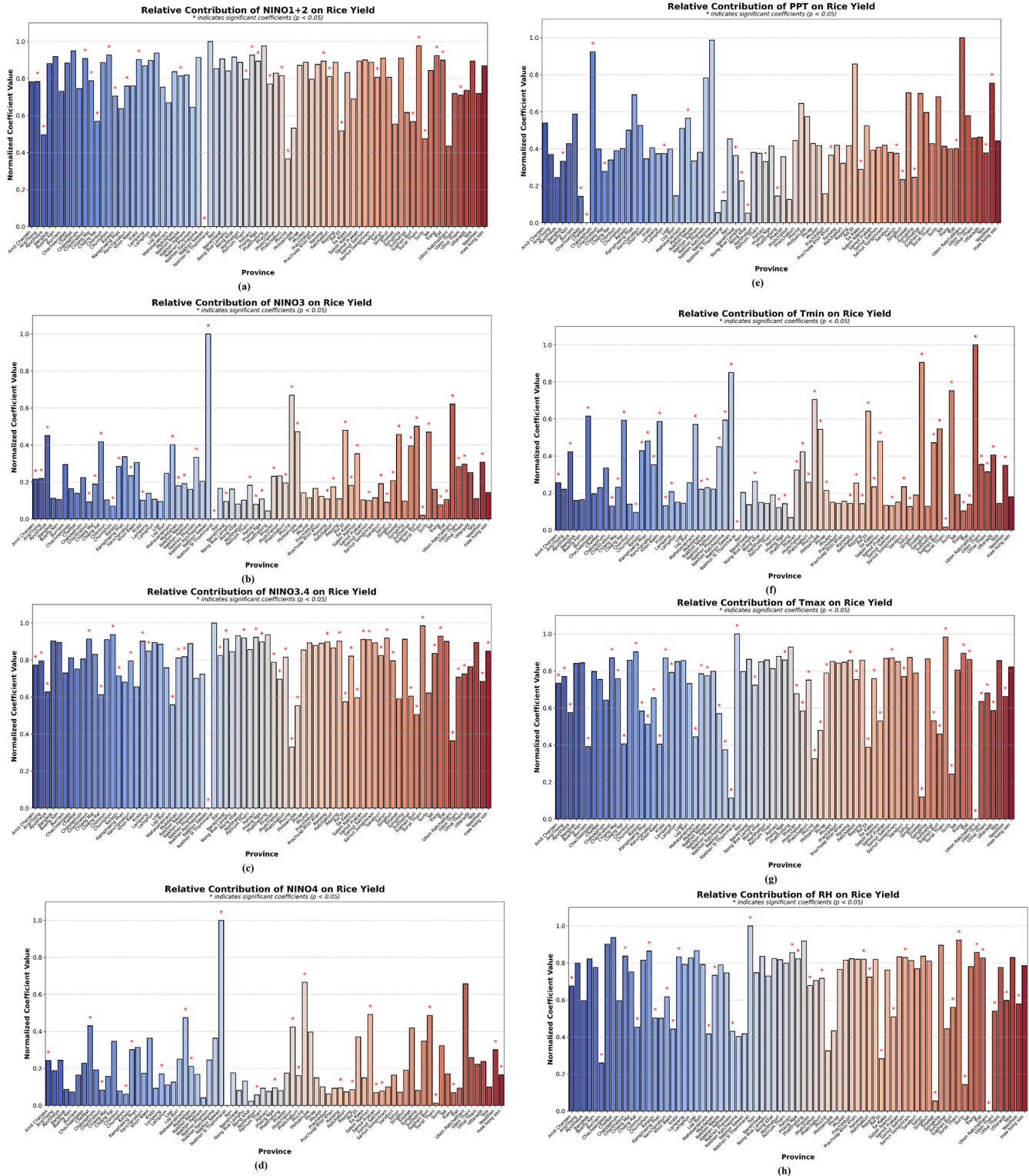


Fig. 5. Relative contributions of ENSO indices and climate variables to rainfed rice yields across provinces in Thailand, as determined by MLR-VIF analysis: (a) Nino 1 + 2, (b) Nino 3, (c) Nino 3.4, (d) Nino 4, (e) rainfall, (f) minimum temperature (g) maximum temperature and (h) relative humidity (RH).

= 0.751). Surat Thani demonstrates a strong model fit ($R^2 = 0.802$). NINO3.4 significantly reduces yields (p-value < 0.001), while $T_{\max}$ and $T_{\min}$ have adverse effects (p-values < 0.001). PPT positively influences yields (p-value < 0.01), highlighting water availability's mitigating role. In Nakhon Sawan, the model explains 70 % of yield variance ($R^2 = 0.699$). ENSO indices NINO3 and NINO3.4 significantly reduce yields (p-values < 0.001), while $T_{\max}$ and $T_{\min}$ negatively affect yields (p-values < 0.01). PPT remains non-significant (p-value = 0.342). Phang Nga shows high model accuracy ($R^2 = 0.776$). ENSO and $T_{\max}/T_{\min}$ negatively influence yields (p-values < 0.001), while PPT is non-significant (p-value = 0.547). For Roi Et ($R^2 = 0.706$) and Yasothorn ($R^2 = 0.691$), ENSO and temperature significantly reduce yields (p-values < 0.05). PPT positively impacts (p-values < 0.05), emphasizing rainfall's role. In contrast, Samut Songkhram ($R^2 = 0.203$) shows no significant predictors due to multicollinearity (e.g., NINO3.4, VIF = 73.05), while Phetchaburi ($R^2 = 0.782$) mirrors other provinces with significant ENSO and temperature impacts (p-values < 0.001).

The VIF analysis indicates the presence of multicollinearity among the predictors (Table S3). A VIF < 10 indicates significant multicollinearity, which can distort the interpretation of regression coefficients. In this study, the VIF values for ENSO indices (NINO1+2, NINO3, NINO3.4, NINO4) and temperature variables ($T_{\max}$, $T_{\min}$) are notably high across most provinces, often exceeding 10. For example, in Bueng Kan, the VIF for NINO3 is 342.80, and for $T_{\min}$, it is 184.47, suggesting severe multicollinearity. This high multicollinearity implies that these variables are highly correlated, making isolating their individual effects on rice yield challenging. Such high VIF values underscore caution in interpreting the regression coefficients, as the estimates may be unstable or unreliable. In contrast, PPT and R_H exhibit lower VIF values, often below 5, indicating less multicollinearity. For example, in Amit Charoen, the VIF for PPT is 1.71, and for R_H , it is 4.24. It implies that these variables have a low correlation with others and can be explained more accurately. The lower multicollinearity for PPT and R_H is a more promising indicator of rice yield in places with significant p-values. The regression coefficients show the nature and strength of the association of the predictors with the rice yield. ENSO indices that reflect the oceanic temperature anomalies in various parts of the Pacific are significant in varying degrees in provinces. For example, in Chanthaburi, NINO3 and NINO3.4 have a significant negative coefficient ($p < 0.001$), meaning that warmer oceanic conditions in these regions are inversely related to rice production. It correlates with the effects of El Niño events, which cause droughts and low rainfall in SEA, which is unsuitable for rice production. Likewise, in Chiang Rai, both NINO3 and NINO3.4 correlate negatively with rice yield at $p < 0.01$, which supports the impact of El Niño on crop yield. However, in some provinces, including Bangkok, the ENSO indices are insignificant ($p > 0.05$), implying that other climate factors may be more influential in determining rice yield in these provinces. Some local climate variables, such as rainfall and temperature, have a positive relationship with rice yield in some provinces. For example, in Kanchanaburi province, the coefficients of $T_{\max}$ and $T_{\min}$ are negative and significant at 0.001, meaning that hot temperatures result in low rice yields. It aligns with the physiological stress that hot temperatures cause to the rice plant and at the most sensitive developmental phases, such as flowering and grain filling.

Rainfall also shows significant positive relationships in some provinces, such as Surat Thani (p-value < 0.01), where increased rainfall is associated with higher rice yields. It reflects the importance of adequate water availability for rice cultivation, particularly in rainfed systems. However, in other provinces, such as Bangkok, PPT is insignificant, highlighting the variability in water availability and irrigation infrastructure across regions. The R-squared values measure how well the regression models explain the variability in rice yield. Provinces such as Bueng Kan ($R^2 = 0.905$) and Surat Thani ($R^2 = 0.802$) exhibit high R-squared values, indicating that the models explain a substantial proportion of the variance in rice yield. These high values suggest that the

selected predictors, including ENSO indices and local climate variables, are strongly associated with rice yield in these regions. The significant coefficients for ENSO indices and local climate variables in these provinces further support their importance in explaining yield variability.

Conversely, provinces such as Samut Songkhram ($R^2 = 0.203$) and Tak ($R^2 = 0.326$) have low R-squared values, indicating that the models explain only a small proportion of the variance in rice yield. Additionally, the high multicollinearity among predictors in these provinces may have contributed to the poor model performance. The analysis demonstrates that ENSO-induced climate signals, particularly NINO3 and NINO3.4, significantly impact rice yield in many Thai provinces, often exerting a negative influence due to their association with drought conditions. Local climate variables, such as temperature and rainfall, also play a critical role, with higher temperatures reducing yields and increased precipitation enhancing them. However, the high multicollinearity among ENSO indices and temperature variables complicates the interpretation of their individual effects.

3.2. Prediction of ENSO-induced impact on rainfed rice yield

The ENSO—CropNet model's performance in rice crop yield in Thailand is highlighted by the deep DNN architecture of the model, which is intended to capture non-linear relationships inherent in climatic and yield data (Fig. 6). With an R^2 of 0.89, the model can explain much of the variability in rice yield, which is not easy given the complexity of climatic factors and large-scale phenomena such as ENSO (Fig. 6(a)). Fig. 6(c) shows the MAE of 1.04 and RMSE of 1.45, indicating that the model's prediction is almost accurate to the actual yields. However, these low errors suggest that there is still a gap between the predicted and actual values. Such differences could be due to variability in agricultural data and other uncontrolled factors such as nutrient status of soils, pest pressure, farming practices, and randomness of climate shocks. The MSE of 2.1 also measures the average of the squared difference, underlining the importance of model recalibration to reduce such errors. The results of feature importance demonstrate that certain climatic factors are significant in determining rice yield (Fig. 6(b)). The most important predictors for the model are NINO4, $T_{\min}$, and PPT, with normalized importance of 1.00, 0.96, and 0.91, respectively. It is consistent with agronomic principles because $T_{\min}$ and rainfall are fundamental factors that define crop growth and development. The importance of these features when log-scaled also highlights the dominance of these features, with NINO4 and $T_{\max}$ being key features. The actual and predicted crop yield for each province is presented in Fig. 7.

The model's effectiveness in capturing such dependencies is essential for accurately predicting yield in climate variability and change. The results of the ENSO—CropNet model show that it has a high capacity for rice yield prediction and uses the most modern DNN methods to identify non-linear dependencies. Although the model is sound, there is a need for continuous updating and calibration to remove the remaining level of uncertainty and improve the model's accuracy due to changing climatic conditions and farming practices.

The analysis of the results of using the RF-XGBoost model to predict the yield of rice crops is presented in Fig. 8. The R^2 of 0.82 shows that the model can capture the various interactions between climatic factors, lagged yield, and rice yield (Fig. 8(a)). This high explanatory power could be attributed to the fact that the current model comprises RF, which is known to be less sensitive to the type of data and capable of minimizing overfitting while incorporating the sequential error correction feature of XGBoost. The hyperparameters, such as 300 n_estimators for RF, max_depth of 20 for RF, and a learning rate of 0.1 with 300 n_estimators for XGBoost, enhanced the model's predictive accuracy. These settings control the complexity of the model and its ability to generalize, thus avoiding overfitting while achieving high predictive accuracy.

However, we have seen that the MAE of 3.62 and RMSE of 3.84 suggest room for improvement (Fig. 8(c)). These error metrics imply



Fig. 6. Performance of ENSO—CropNet in predicting rainfed rice yields: (a) Parity plot comparing actual versus predicted yields, (b) Feature importance ranking for yield prediction, and (c) Comparison of actual and predicted yields across provinces in Thailand.

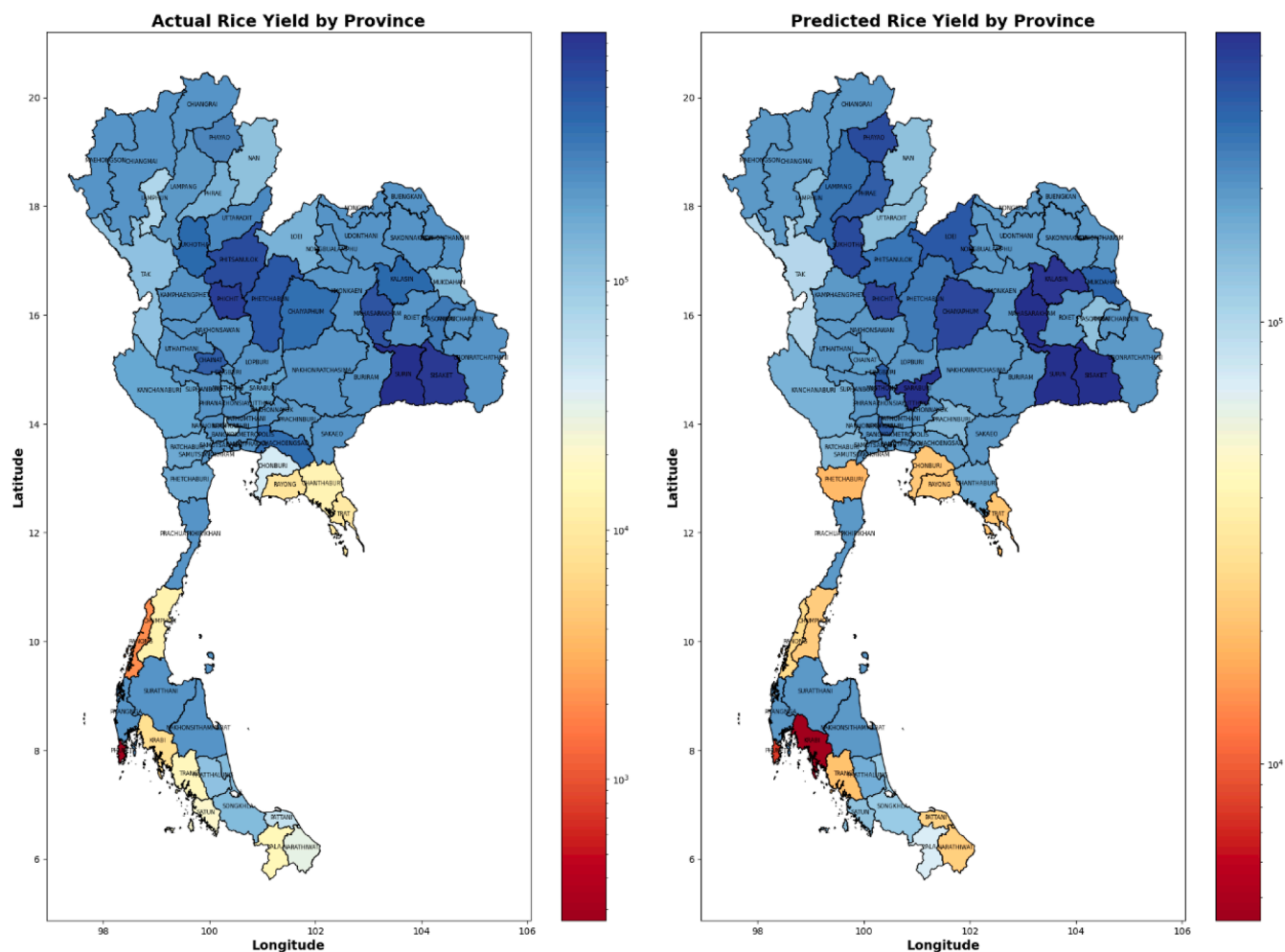


Fig. 7. Actual vs. Predicted rice yield by province using the ENSO—CropNet model.

that although the model is relatively accurate, it is not accurate for extreme yield values, as observed from the parity plot at the higher end of the yield scale. It could be due to non-linear interactions or data points the model does not capture well. Further, the moderate MSE of 1.71 shows that although the model prediction is reasonably accurate in most cases, there are cases where prediction errors are significant in areas with extreme climate or yield fluctuations.

The SHAP analysis reveals the model interpretability and highlights rainfall, temperature, and lagged yield as the most important features (Fig. 8(b)). It conforms to agronomic factors, which are well understood to have a profound influence on the growth and yield of rice. However, the model's reliance on these variables also suggests that it may be less effective in regions where other factors, such as soil quality or pest incidence, play a more critical role. Incorporating these additional features could enhance the model's predictive accuracy and generalizability.

The results of this study contribute valuable knowledge of ENSO effects on rainfed rice yields in Thailand, using sophisticated AI methods for yield fluctuation estimation and forecasting. The study reveals that ENSO climate signals, especially Niño3 and Niño3.4 indices, affect rice yields and that during El Niño years, when they induce drought stress, yields are low, but during La Niña years, when there is increased rainfall, there are benefits in terms of yield increases but with costs like flooding and pest attacks. These findings agree with other studies that reveal that El Niño affects crop productivity in Southeast Asia, particularly the rainfed systems (M. Waqas et al., 2024; Kirono and Tapper, 1999). However, this study extends this literature by employing DL models such

as ENSO—CropNet to capture the non-linear relationship between ENSO indices, local climate predictors, and rice yields.

The ENSO—CropNet model had an R^2 of 0.89. It indicates that the model can explain much of the total variance in rice yields. This high explanatory power is due to the ability of the model to capture local and global dependencies of the data, which cannot be done in statistical models due to their linearity (Wang et al., 2023; B. Liu et al., 2017). The MAE and RMSE values of 1.04 and 1.45, respectively, prove the predictive accuracy of the model; however, the MSE of 2.1 shows that there is still room for improvement in the high yield. Such differences may result from differences in soil nutrient status, pests and diseases, and other management practices not captured in the model. The results of the feature importance analysis revealed that Niño4, T_{min} , and PPT were the most important predictors, which is consistent with agronomic studies that highlight the role of temperature and water in crop production (Lobell et al., 2006; Peng et al., 2004).

The ENSO—CropNet results also compared the RF-XGBoost ensemble model with the other models, and the authors found that the model also had a good predictive performance with R^2 of 0.82. However, it has a higher MAE of 3.62 and RMSE of 3.84, which indicates that ENSO—CropNet is better than the model regarding yield prediction, especially for the outliers. It is because the model employs basic decision tree structures, and although they are effective in handling high-dimensional data, they do not fully account for the complex interactions between ENSO phases and local climate parameters (Breiman, 2001; Chen and Guestrin, 2016). The feature's contribution to the model RF-XGBoost was demonstrated in the SHAP analysis, where the most key

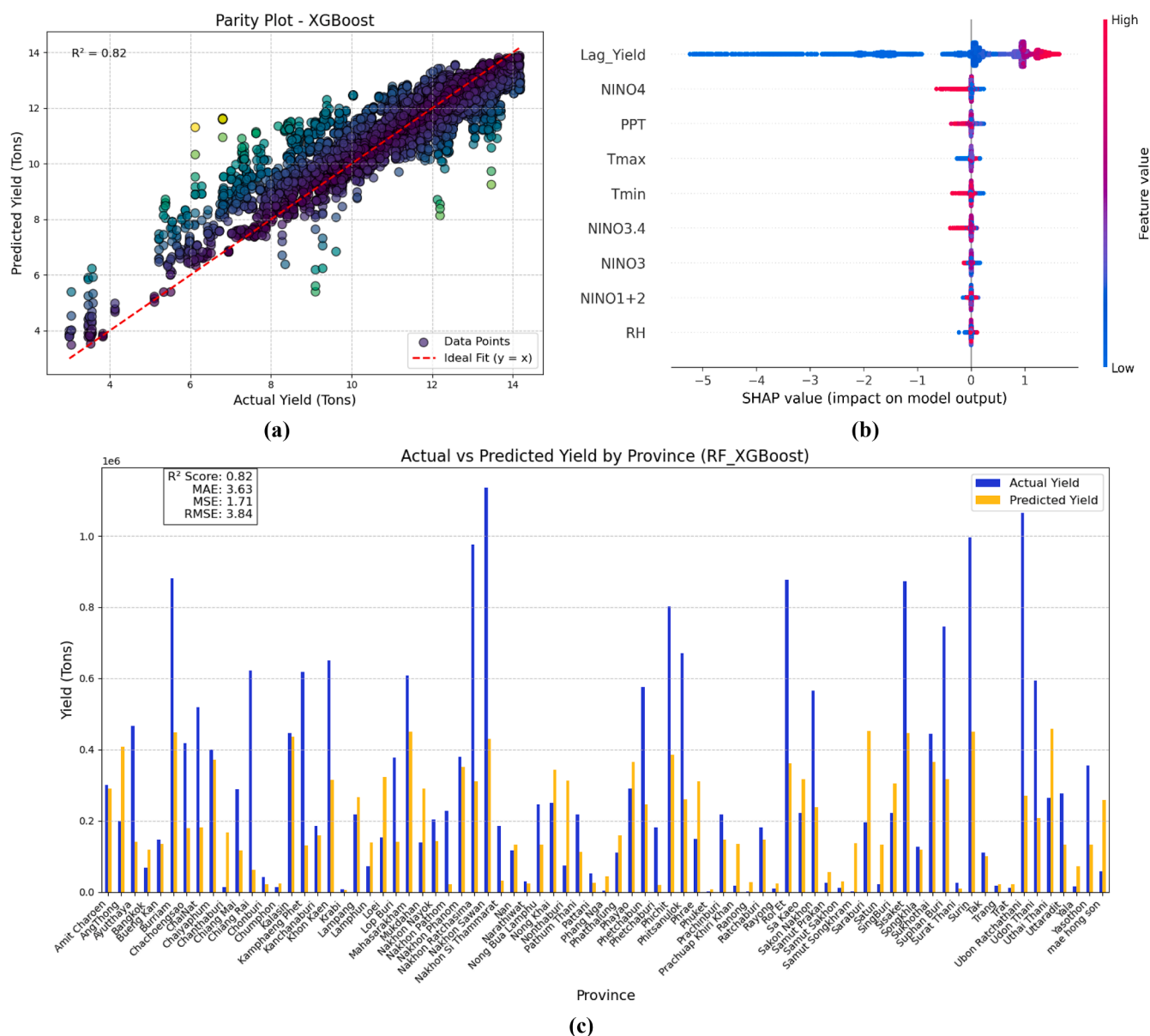


Fig. 8. Performance of RF-XGBoost in predicting rainfed rice yields: (a) Parity plot comparing actual versus predicted yields, (b) SHAP values for feature importance for yield prediction, and (c) Comparison of actual and predicted yields across provinces in Thailand.

features were rainfall, temperature, and lagged yield, such as ENSO—CropNet.

The results align with previous research that has established that El Niño reduces rice production in Thailand and other Southeast Asian nations. For example, Wongkhunkaew et al. (2020) noted that during the El Niño episodes, drought negatively affected rice yields in the rainfed farming systems of northeastern Thailand (Wongkhunkaew et al., 2020). Likewise, Felkner et al. (2009) reported that El Niño caused significant yield reductions in the central irrigated zones of Thailand; however, the losses were lower than in rainfed zones because of irrigation (Felkner et al., 2009). This study expands on these findings by presenting a more detailed picture of ENSO effects on yield across different provinces and the prominent level of provincial heterogeneity in yield responses to ENSO events. For example, in Chiang Mai, the model could explain 47 % of the yield variance, Niño3 and Niño3.4 having negative coefficients with yield. In Surat Thani, the model explained 80 % of the variance, Niño3.4 having a substantial negative impact on the yield. These variations indicate that climate risks related to ENSO require a regional perspective in their mitigation.

This study's contribution is applying AI-based techniques to identify non-linear relationships between ENSO events and rice yields. ENSO—CropNet model, with the feature representation hierarchy and the spatiotemporal structure learning capability, is a more reliable method of determining yield fluctuation under ENSO climate conditions. The model was designed to have five hidden layers, each of which is connected to the next layer with all neurons, and the number of neurons in each successive layer is reduced; batch normalization; and dropout layers, which were designed to solve the issues of overfitting and vanishing gradients in DL applications (Ioffe, 2015; Srivastava et al., 2014). Along with the selection of the Adam optimizer and the learning rate schedule explicitly designed for this task, it provided high predictive accuracy and model stability.

However, the following limitations are worth noting in the study: The VIF values above 10 in most provinces of the study area indicate high multicollinearity between ENSO indices and temperature variables, making it challenging to determine the effect of each on rice yields. This problem is even more evident in provinces such as Bueng Kan, where the VIF for Niño3 was 342.80, which implies that the regression coefficients

could be unreliable or erroneous (O'Brien, 2007).

4. Conclusion and recommendations

ENSO, originating from the tropical Pacific Ocean, exerts significant global influence, often causing extreme climatic events such as severe droughts and floods. It impacted rice yield in many regions, including Thailand. This study quantified and predicted the impact of ENSO-induced climate signals on rainfed rice yields in Thailand using advanced AI-based methodologies, offering novel insights into the complex interplay between large-scale climatic phenomena and agricultural productivity. The integration of DNNs in ENSO—CropNet and ensemble models like RF-XGBoost represents a significant advancement in capturing the non-linear and spatiotemporal dependencies inherent in climate-yield relationships. The novelty of this research lies in its province-specific analysis, which accounts for regional climatic variability and agricultural practices, and its application of hierarchical feature representation to disentangle the multifaceted impacts of ENSO on rice production.

The findings revealed that ENSO phases, particularly El Niño and La Niña, influence rice yields by modulating temperature and precipitation patterns. ENSO-induced droughts during El Niño events were associated with significant yield reductions, particularly in rainfed systems in northeastern Thailand, where water availability is critical. Conversely, La Niña-related excessive rainfall, while beneficial in some regions, posed challenges such as flooding and pest outbreaks. The MLR analysis highlighted the negative correlations between ENSO indices (e.g., NINO3, NINO3.4) and yields, with temperature variables ($T_{\max}$, $T_{\min}$) exacerbating these effects. However, high multicollinearity among predictors underscored the complexity of isolating individual climatic drivers.

Despite the robust performance of AI models (ENSO—CropNet: $R^2 = 0.89$; RF-XGBoost: $R^2 = 0.82$), limitations include the models' reduced accuracy in predicting extreme yield values and their reliance on historical data, which may not fully capture future climate variability. Notably, the study found that ENSO-driven climate variability led to a 12 % decline in rice yields across northern provinces. The findings underscore the significant role of ENSO-induced climate variability in rainfed rice production, with AI models such as ENSO—CropNet offering highly accurate predictions.

Additionally, the study did not account for non-climatic factors such as soil fertility and pest dynamics, which could further refine predictions. Recommendations include expanding datasets to incorporate soil and pest variables, refining model architectures to capture extreme events better, and integrating real-time climate forecasts for initiative-taking agricultural planning. This study underscores the potential of AI-driven approaches to enhance climate resilience in rice production, offering actionable strategies to mitigate ENSO-induced risks and support sustainable food security in Thailand.

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CRedit authorship contribution statement

Usa Humphries Wannasingha: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Muhammad Waqas:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shakeel Ahmad:** Writing – review & editing, Visualization, Validation. **Angkool Wangwongchai:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Porntip Dechpichai:** Visualization, Validation, Software, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

References

- Abid, M.A., Almazroui, M., Kucharski, F., O'Brien, E., Yousef, A.E., 2018. ENSO relationship to summer rainfall variability and its potential predictability over Arabian Peninsula region. *NPJ. Clim. Atmos. Sci.* 1 (1), 20171.
- Adams, R.M., Chen, C.-C., McCarl, B.A., Weiher, R.F., 1999. The economic consequences of ENSO events for agriculture. *Clim. Res.* 13 (3), 165–172.
- Ahmad, J., Alam, D., Haseen, M.S., 2011. Impact of climate change on agriculture and food security in India. *Int. J. Agricult. Environ. Biotechnol.* 4 (2), 129–137.
- A.Q. Al-Amin, M.M. Alam, and B. Doberstein, "COP adoption, implications and projections of climate change mitigation policy for Asean Region," 2021.
- Allen, J.T., Karoly, D.J., 2014. A climatology of Australian severe thunderstorm environments 1979–2011: inter-annual variability and ENSO influence. *Int. J. Climatol.* 34 (1).
- An, S.I., Tziperman, E., Okumura, Y.M., Li, T., 2020. ENSO irregularity and asymmetry. *El Niño Southern Oscillat. Chang. Climate* 153–172.
- Angulo, C., Becker, M., Wassmann, R., 2012. Yield gap analysis and assessment of climate-induced yield trends of irrigated rice in selected provinces of the Philippines. *J. Agricult. Rural Devel. Trop. Subtrop.* 113 (1), 61–68.
- Aslan, M.F., Sabanci, K., Aslan, B., 2024. Artificial intelligence techniques in crop yield estimation based on Sentinel-2 data: a comprehensive survey. *Sustainability*. 16 (18), 8277.
- Assimakopoulos, F., Vassilakis, C., Margaritis, D., Kotis, K., Spiliotopoulos, D., 2024. Artificial intelligence tools for the agriculture value chain: status and prospects. *Electronics* 13 (22), 4362.
- Basso, B., Liu, L., 2019. Seasonal crop yield forecast: methods, applications, and accuracies. *Adv. Agronomy* 154, 201–255.
- Battisti, D., Sarachik, E., 1995. Understanding and predicting ENSO. *Rev. Geophys.* 33 (S2), 1367–1376.
- Bhatti, U.A., et al., 2021. Local similarity-based spatial-spectral fusion hyperspectral image classification with deep CNN and Gabor filtering. *IEEE Trans. Geosci. Remote Sens.* 60, 1–15.
- Boer, R., Surmaini, E., 2020. Economic benefits of ENSO information in crop management decisions: case study of rice farming in West Java, Indonesia. *Theor. Appl. Climatol.* 139, 1435–1446.
- Breiman, L., 2001. Random forests. *Mach. Learn.* 45, 5–32.
- Cai, J., et al., 2024. A novel graph-attention based multimodal fusion network for joint classification of hyperspectral image and LiDAR data. *Expert Syst. Appl.* 249, 123587.
- Chambers, J.M., Hastie, T.J., 2017. Statistical models. *Statistical Models in S*: Routledge, pp. 13–44.
- CHANALERT, W., Janjai, S., 2022. Investigation of the Expansion of the Tropical Zone and Its Effect On Climate in Thailand. *Silpakorn University*.
- Chen, T., Guestrin, C., 2016. Xgboost: a scalable tree boosting system. In: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794.
- Ciaburro, G., Venkateswaran, B., 2017. *Neural Networks with R: Smart models Using CNN, RNN, Deep learning, and Artificial Intelligence Principles*. Packt Publishing Ltd.
- Cirino, P.H., Féres, J.G., Braga, M.J., Reis, E., 2015. Assessing the impacts of ENSO-related weather effects on the Brazilian agriculture. *Proc. Econom. Finance* 24, 146–155.
- Dang, V.H., et al., 2020. Assessment of rainfall distributions and characteristics in coastal provinces of the Vietnamese Mekong Delta under climate change and ENSO processes. *Water* 12 (6), 1555.
- Dangthaisong, P.; et al., "Differences in growing region affected grain yield, grain quality and 2-acetyl-1-pyrroline content in Thai jasmine rice (*Oryza sativa* L.).

- Devaraj, J., Madurai Elavarasan, R., Shafiullah, G., Jamal, T., Khan, I., 2021. A holistic review on energy forecasting using big data and deep learning models. *Int. J. Energy Res.* 45 (9), 13489–13530.
- Dhanke, J.A., et al., 2024. Climate-based AI-powered precision irrigation: sustainably smart agriculture frameworks for maximum crop yields. *Remote Sens. Earth. Syst. Sci.* 1–12.
- Fang, W., Sha, Y., Sheng, V.S., 2022. Survey on the application of artificial intelligence in ENSO forecasting. *Mathematics* 10 (20), 3793.
- Felkner, J., Tazhibayeva, K., Townsend, R., 2009. Impact of climate change on rice production in Thailand. *Am. Econom. Rev.* 99 (2), 205–210.
- Fischer, G., Shah, M., Tubiello, F.N., Van Velhuizen, H., 2005. Socio-economic and climate change impacts on agriculture: an integrated assessment, 1990–2080. *Philosoph. Trans. R. Soc. B Biol. Sci.* 360 (1463), 2067–2083.
- Friedman, J.H., 2001. Greedy function approximation: a gradient boosting machine. *Ann. Stat.* 1189–1232.
- Gandhi, N., Armstrong, L.J., Petkar, O., Tripathy, A.K., 2016. Rice crop yield prediction in India using support vector machines. In: 2016 13th International Joint Conference on Computer Science and Software Engineering (JCSSE). IEEE, pp. 1–5.
- Gariano, S.L., Guzzetti, F., 2016. Landslides in a changing climate. *Earth. Sci. Rev.* 162, 227–252.
- Groß, J., 2003. Variance inflation factors. *R news* 3 (1), 13–15.
- Grubbs, F.E., Beck, G., 1972. Extension of sample sizes and percentage points for significance tests of outlying observations. *Technometrics*. 14 (4), 847–854.
- Hammer, G., et al., 2001. Advances in application of climate prediction in agriculture. *Agric. Syst.* 70 (2–3), 515–553.
- Hou, J., et al., 2024. A comprehensive evaluation of influencing factors of neonicotinoid insecticides (NEOs) in farmland soils across China: first focus on film mulching. *J. Hazard. Mater.* 470, 134284.
- Ioffe, S., "Batch normalization: accelerating deep network training by reducing internal covariate shift," *arXiv preprint arXiv:1502.03167*, 2015.
- Jintrawet, A., Buddhaboon, C., 2011. El Niño–Southern oscillation and rice production in Thailand during 1980–2002 period. In: International conference on the role of agriculture and natural resources on global changes (ANGC2011), Chiang Mai.
- Kattenborn, T., Leitloff, J., Schiefer, F., Hinz, S., 2021. Review on Convolutional Neural Networks (CNN) in vegetation remote sensing. *ISPRS J. Photogr. Remote Sens.* 173, 24–49.
- Kingma, D.P.; "Adam: a method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- Kirono, D.G., Tapper, N.J., 1999. ENSO rainfall variability and impacts on crop production in Indonesia. *Phys. Geogr.* 20 (6), 508–519.
- Kourou, K., Exarchos, T.P., Exarchos, K.P., Karamouzis, M.V., Fotiadis, D.I., 2015. Machine learning applications in cancer prognosis and prediction. *Comput. Struct. Biotechnol.* 13, 8–17.
- Kumar, S., Anusha, B., Bhatia, A., 2024. Soil, water, and crop management practices to mitigate greenhouse gases emission. *Climate Change Soil-Water-Plant Nexus: Agricult. Environ.* 189–222.
- Kyaw, Y., Nguyen, T.P.L., Winijkul, E., Xue, W., Virdis, S.G., 2023. The effect of climate variability on cultivated crops' Yield and farm income in Chiang Mai Province, Thailand. *climate* 11 (10), 204.
- Lasco, R.D., Habito, C.M.D., Delfino, R.J.P., Pulhin, F.B., Concepcion, R.N., 2011. Climate change adaptation for smallholder farmers in Southeast Asia.
- Lee, S., 2021. In the era of climate change: moving beyond conventional agriculture in Thailand. *Asian J. Agricult. Devel.* 18 (1), 1–14.
- Li, S., Song, W., Fang, L., Chen, Y., Ghamisi, P., Benediktsson, J.A., 2019. Deep learning for hyperspectral image classification: an overview. *IEEE Trans. Geosci. Remote Sens.* 57 (9), 6690–6709.
- Liaw, A., 2002. Classification and Regression By Randomforest. *R news*.
- Liu, Y., Yang, X., Wang, E., Xue, C., 2014. Climate and crop yields impacted by ENSO episodes on the North China Plain: 1956–2006. *Reg. Environ. Change* 14, 49–59.
- Liu, B., Wei, Y., Zhang, Y., Yang, Q., 2017a. Deep neural networks for high dimension, low sample size data. *IJCAI* 2017, 2287–2293.
- Liu, W., Wang, Z., Liu, X., Zeng, N., Liu, Y., Alsaadi, F.E., 2017b. A survey of deep neural network architectures and their applications. *Neurocomputing*. 234, 11–26.
- Lobell, D.B., Field, C.B., Cahill, K.N., Bonfils, C., 2006. Impacts of future climate change on California perennial crop yields: model projections with climate and crop uncertainties. *Agric. For. Meteorol.* 141 (2–4), 208–218.
- Lu, J., et al., 2024. Deep learning for multi-source data-driven crop yield prediction in Northeast China. *Agriculture* 14 (6), 794.
- Lundberg, S.; "A unified approach to interpreting model predictions," *arXiv preprint arXiv:1705.07874*, 2017.
- Manjunath, M.C., Palayyan, B.P., 2023. An efficient crop yield prediction framework using hybrid machine learning model. *Revue d'Intelligence Artificielle* 37 (4).
- Masud, M.B., Soni, P., Shrestha, S., Tripathi, N.K., 2016. Changes in climate extremes over North Thailand, 1960–2099. *J. Climatol.* 2016 (1), 4289454.
- Molle, F.; "Agrarian change in the Central Plain of Thailand (1950–2020): a longitudinal study of farm characteristics," 2024.
- Mythili, K., Rangaraj, R., 2021. Deep learning with particle swarm based hyper parameter tuning based crop recommendation for better crop yield for precision agriculture. *Ind. J. Sci. Technol.* 14 (17), 1325–1337.
- Naylor, R.L., Battisti, D.S., Vimont, D.J., Falcon, W.P., Burke, M.B., 2007. Assessing risks of climate variability and climate change for Indonesian rice agriculture. *Proc. Natl. Acad. Sci.* 104 (19), 7752–7757.
- Ngaira, J.K.W., 2007. Impact of climate change on agriculture in Africa by. *Sci. Res. Essays* 2 (7), 238–243.
- Nounmusig, W., 2018. Analysis of rainfall in the eastern Thailand. *GEOMATE J.* 14 (46), 150–155.
- O'Brien, R.M., 2007. A caution regarding rules of thumb for variance inflation factors. *Qual. Quant.* 41, 673–690.
- Oikonomidis, A., Catal, C., Kassahun, A., 2023. Deep learning for crop yield prediction: a systematic literature review. *N. Z. J. Crop Hortic. Sci.* 51 (1), 1–26.
- Parry, M.L., Rosenzweig, C., Iglesias, A., Livermore, M., Fischer, G., 2004. Effects of climate change on global food production under SRES emissions and socio-economic scenarios. *Global Environ. Change* 14 (1), 53–67.
- Paz, J.O., Woli, P., y Garcia, A.G., Hoogenboom, G., 2012. Cotton yields as influenced by ENSO at different planting dates and spatial aggregation levels. *Agric. Syst.* 111, 45–52.
- Peng, S., et al., 2004. Rice yields decline with higher night temperature from global warming. *Proc. Natl. Acad. Sci.* 101 (27), 9971–9975.
- Pipitukdee, S., Attavanich, W., Bejranonda, S., 2020. Impact of climate change on land use, yield and production of cassava in Thailand. *Agriculture* 10 (9), 402.
- Potgieter, A.B., Hammer, G.L., Meinke, H., Stone, R.C., Goddard, L., 2005. Three putative types of El Niño revealed by spatial variability in impact on Australian wheat yield. *J. Clim.* 18 (10), 1566–1574.
- Reda, A.G., Tripathi, N.K., Soni, P., Tipdecho, T., 2015. Rain fed rice agriculture under climate variability in Southeast Asia: the case of Thailand. *J. Earth Sci. Clim. Change* 6 (8), 1.
- Reddy, M.P.; Mathur, A.K.; Jain, R.K.; Agarwal, S.K.; and Singh, S.; "Climate change and weather variability in crop modelling: evidence from rice yield trials in India using LSTM model," 2022.
- Rembold, F., Atzberger, C., Savin, I., Rojas, O., 2013. Using low resolution satellite imagery for yield prediction and yield anomaly detection. *Remote Sens.* 5 (4), 1704–1733.
- Rodríguez Álvarez, L.; "Chinese securitization of the main water sources in South and Southeast Asia: brahmaputra and Mekong comparison," 2024.
- Rodysill, J.R., Russell, J.M., Vuille, M., Dee, S., Lunghino, B., Bijaksana, S., 2019. La Niña-driven flooding in the Indo-Pacific warm pool during the past millennium. *Quat. Sci. Rev.* 225, 106020.
- Saha, M., Nanjundiah, R.S., 2020. Prediction of the ENSO and EQUINOX indices during June–September using a deep learning method. *Meteorol. Appl.* 27 (1), e1826.
- Salvador, C., Nieto, R., Linares, C., Díaz, J., Gimeno, L., 2020. Effects of droughts on health: diagnosis, repercussion, and adaptation in vulnerable regions under climate change. *Challenges for future research. Sci. Total Environ.* 703, 134912.
- Sazib, N., Mladenova, L.E., Bolten, J.D., 2020. Assessing the impact of ENSO on agriculture over Africa using earth observation data. *Front. Sustain. Food Syst.* 4, 509914.
- Shuai, J., Zhang, Z., Tao, F., Shi, P., 2016. How ENSO affects maize yields in China: understanding the impact mechanisms using a process-based crop model. *Int. J. Climatol.* 36 (1).
- Shukla, S., et al., 2020. Improving early warning of drought-driven food insecurity in southern Africa using operational hydrological monitoring and forecasting products. *Natural Hazards Earth Syst. Sci.* 20 (4), 1187–1201.
- Singhtrattana, N., 2003. Interannual and Interdecadal Variability of Thailand summer monsoon: Diagnostic and Forecast. University of Colorado.
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15 (1), 1929–1958.
- Su, T.-L., Jaki, T., Hickey, G.L., Buchan, I., Sperrin, M., 2018. A review of statistical updating methods for clinical prediction models. *Stat Methods Med. Res.* 27 (1), 185–197.
- Tao, F., Yokozawa, M., Xu, Y., Hayashi, Y., Zhang, Z., 2006. Climate changes and trends in phenology and yields of field crops in China, 1981–2000. *Agric. For. Meteorol.* 138 (1–4), 82–92.
- Tranmer, M., Elliot, M., 2008. Multiple linear regression. *Cathie Marsh Centre for Census Survey Res. (CCSR)* 5 (5), 1–5.
- Vaulet, T., et al., 2022. Gradient boosted trees with individual explanations: an alternative to logistic regression for viability prediction in the first trimester of pregnancy. *Comput. Methods Programs Biomed.* 213, 106520.
- Wang, C., Picaut, J., 2004. Understanding ENSO physics—A review. *Earth's Climate: The Ocean–Atmosphere Int. Geophys. Monogr.* 147, 21–48.
- Wang, B., Feng, P., Waters, C., Cleverly, J., Liu, D.Li, Yu, Q., 2020. Quantifying the impacts of pre-occurred ENSO signals on wheat yield variation using machine learning in Australia. *Agric. For. Meteorol.* 291, 108043.
- Wang, G.-G., Cheng, H., Zhang, Y., Yu, H., 2023. ENSO analysis and prediction using deep learning: a review. *Neurocomputing* 520, 216–229.
- Wangwongchai, A., Waqas, M., Dechpichai, P., Hlaing, P.T., Ahmad, S., Humphries, U. W., 2023. Imputation of missing daily rainfall data: A comparison between artificial intelligence and statistical techniques. *MethodsX*. 11, 102459.
- Waqas, M., Bonnet, S., Wannasing, U.H., Hlaing, P.T., Lin, H.A., Hashim, S., 2023a. Assessment of advanced artificial intelligence techniques for flood forecasting. In: 2023 International Multi-disciplinary Conference in Emerging Research Trends (IMCERT), 1. IEEE, pp. 1–6.
- Waqas, M., Humphries, U.W., Wangwongchai, A., Dechpichai, P., Ahmad, S., 2023b. Potential of artificial intelligence-based techniques for rainfall forecasting in Thailand: a comprehensive review. *Water* 15 (16), 2979.
- Waqas, M., Naseem, A., Humphries, U.W., Hlaing, P.T., Shoaib, M., Hashim, S., 2024a. A comprehensive review of the impacts of climate change on agriculture in Thailand. *Farming Syst.* 100114.
- Waqas, M., Humphries, U.W., Hlaing, P.T., 2024b. Time series trend analysis and forecasting of climate variability using deep learning in Thailand. *Results. Eng.* 24, 102997.

- Waqas, M., Humphries, U.W., Hlaing, P.T., Ahmad, S., 2024c. Seasonal WaveNet-LSTM: a deep learning framework for precipitation forecasting with integrated large scale climate drivers. *Water* 16 (22), 3194.
- Waqas, M., Humphries, U.W., Chueasa, B., Wangwongchai, A., 2024d. Artificial intelligence and numerical weather prediction models: a technical survey. *Natural Hazards Res.*
- Waqas, M., Humphries, U.W., Hlaing, P.T., Wangwongchai, A., Dechpichai, P., 2024e. Advancements in daily precipitation forecasting: a deep dive into daily precipitation forecasting hybrid methods in the tropical climate of Thailand. *MethodsX*. 12, 102757.
- Waqas, M., Humphries, U.W., Wangwongchai, A., Dechpichai, P., Zarin, R., Hlaing, P.T., 2024f. Incorporating novel input variable selection method for in the different water basins of Thailand. *Alexandr. Eng. J.* 86, 557–576.
- Watanabe, K., 2017. Improvement in rainfed rice production during an era of rapid national economic growth: a case study of a village in Northeast Thailand. *Southeast Asian Stud.* 6 (2), 293–306.
- Whitfield, S., et al., 2019. Exploring temporality in socio-ecological resilience through experiences of the 2015–16 El Niño across the tropics. *Global Environ. Change* 55, 1–14.
- Wikarnpapraharn, C., Kositsakulchai, E., 2010. Relationship between ENSO and rainfall in the Central Plain of Thailand. *Agricult. Natural Resources* 44 (4), 744–755.
- Wongkhunkaew, P., Konyai, S., Sriboonlue, V., 2020. Effect of climate variability on Rice production of Northeastern Thailand. *GEOMATE J.* 18 (68), 129–134.
- Yokying, P., Promkhambut, A., 2024. Drought, flood, and Rice security in Central Thailand. *AsiaPacific Issues* (163).
- Yuan, C., Yamagata, T., 2015. Impacts of IOD, ENSO and ENSO Modoki on the Australian winter wheat yields in recent decades. *Sci. Rep.* 5 (1), 17252.
- Yuan, M., et al., 2024. Multi-parameter prediction of solar greenhouse environment based on multi-source data fusion and deep learning. *Agriculture* 14 (8), 1245.
- Zhao, Q., Zhu, Y., Wan, D., Yu, Y., Cheng, X., 2018. Research on the data-driven quality control method of hydrological time series data. *Water* 10 (12), 1712.
- Zhou, L., Pan, S., Wang, J., Vasilakos, A.V., 2017. Machine learning on big data: opportunities and challenges. *Neurocomputing*. 237, 350–361.